Momin Ahmad Khan

momin-ahmad-khan.github.io
Google Scholar profile

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LinkedIn: LinkedIn portfolio

Education

Doctor of Philosophy, Electrical and Computer Engineering Sep. 2021 – Oct. 2026 (Expected)
University of Massachusetts Amherst, USA *CGPA 3.95/4.00*

- PhD student supervised by: Fatima Muhammad Anwar
- Research areas: Mechanistic Interpretability, Federated Learning, Adversarial Machine Learning
- Major courses: Security and Privacy for GenAI, Advanced Information Assurance, Probability and Random Processes

Bachelor of Electrical Engineering Sep. 2017 – Jun. 2021
School of Electrical Engineering and Computer Science (SEECS) *CGPA 3.97/4.00*
National University of Sciences & Technology (NUST), Islamabad, Pakistan

- Undergraduate student supervised by Dr. Hassan Aqeel Khan and Dr. Faisal Shafait
- Final year thesis: "Low-Cost Whole Slide Image Scanner with Deep Learning Applications" (Awarded Rector's Gold medal)
- Major courses: Machine Learning, Computer Vision, Signal Processing, Embedded Systems, Microprocessor Systems

Research Interests

- Mechanistic interpretability of multimodal foundation models via sparse autoencoders and latent-space steering
- Security and robustness in federated learning under data heterogeneity and adversarial manipulation
- Systematic analysis of FL defenses, uncovering evaluation pitfalls and proposing remedies for reliable benchmarking

Experience

Applied Scientist II Aug. 2026 – current
Amazon, Seattle, WA, USA
Team: Amazon Selection and Catalog Systems (ASCS) – Item and Relationship Platform
Research topics: Generative AI, Vision-Language Models, multimodal reasoning, interpretability and uncertainty quantification

- Own the machine learning decision system that reconciles approximately 1 billion daily seller listing submissions against a catalog of approximately 150 billion products, a matching problem without precedent at this scale
- Replacing a legacy rule-based matching system with state-of-the-art multimodal models for product-identity and product-relationship decisions
- Curating high-quality training data and building distillation pipelines that compress frontier multimodal models into production models meeting strict latency constraints
- Developing calibration, uncertainty quantification, and evaluation methodology governing the asymmetric trade-off between false matches and false conflicts
- Safeguarding catalog integrity by preventing harmful, counterfeit, or mislabeled listings from being merged into trusted existing product records

Research Scientist Intern Jun. 2025 – Sep. 2025
Nokia Bell Labs, Murray Hill, NJ, USA
Research topics: Mechanistic Interpretability, Multimodal Foundational Models, Representation Engineering in LLMs/VLMs

- Development of interpretable control pipeline for embodied AI agent (Microsoft MAGMA).
- Exploring Sparse Autoencoders (SAEs) for mechanistic interpretability in multimodal LLMs.
- Proposed inference-time framework for steering multimodal LLMs in the action-space

Research Scientist Intern Jun. 2024 – Aug. 2024
Nokia Bell Labs, Murray Hill, NJ, USA
Research topics: AI agents, image-generation models, speech-to-text and text-to-speech models

- Developed end-to-end autonomous model selection pipeline for in-house Bell Labs infrastructure.
- Used LLama 3-based LLM agents for designing data curation pipelines.
- Prototyped MeetingMate, a smart multimodal meeting assistant, using OpenAI Whisper and Meta ImageBind

Research Assistant and PhD Candidate Sep. 2021 – current
Emerging Embedded Technologies Lab, UMass Amherst, USA
Research topics: Federated Learning, ML Security and Privacy, Interpretable AI

- Designing secure and robust federated learning systems under data heterogeneity and adversarial threats

- Exploring mechanistic interpretability in multimodal foundation models using sparse representations and latent space steering
- Analyzing and improving evaluation practices in FL defenses to enable reliable benchmarking and reproducible research
- Teaching Assistant for Networked Embedded Systems (ECE-635) and Introduction to Embedded Systems (ECE-231)

Undergraduate Research Assistant May 2019 – May 2021
Signal Processing and Machine Learning (SIGMA) Lab in collaboration with TUKL
R&D Center, SEECS, NUST, Islamabad, Pakistan

Main research topics: Medical Imaging, Signal processing, AI on Edge devices, Embedded systems

- Optimization of AI models (YOLOv4, ResNets) with TensorRT on edge devices, including Jetson Nano and Jetson TX2
- Real-time auto-focusing with Laplacian filter as focus metric. Integrated algorithm with stepper motors circuitry
- Segmentation of keratin pearl and epithelium tissues using UNET in Whole Slide images
- Teaching assistant for Digital Logic Design course (EE-221)

Embedded Systems Engineer Aug. 2020 – Dec 2020
Sedenius Technologies (subsidiary of Sedenius Engineering GmbH) and NUST, Pakistan

- Explored Embedded Systems research aspects of Self Driving Cars
- Proposed Embedded and Electronic circuit design for "Mobile Data-logger for Test-bed Scale Car"
- Developed schematics and PCB design for "RC Autonomous Car" in EAGLE

AI Teaching Assistant Jun. 2020 – May 2021
AI Lounge, Islamabad, Pakistan

- Created tailored course content on deep learning and computer vision for school children
- Simplified complex topics using intuitive methods and engaging visuals

Technical Skills

Languages: Python, C/C++, MATLAB, Assembly and embedded C
Programming: PyTorch, Tensorflow, JAX, MXNet, fast.ai, Keras, Latex, TensorRT, Scikit-learn, Hugging-Face, LangChain
Deep Learning: SFT, DPO, GRPO, LLMs, VLMs, VLAs, Classification, Object Detection
Tools: Linux, Slurm, PyCharm, VS code, Git, Google Cloud Platform, AUTOCAD

Publications

DISPEL: Diffusion Interpretability and Selective Patching to Eliminate Latent Backdoors *Under review, ICLR 2027*
***Momin Ahmad Khan**, Yasra Chandio, Fatima Muhammad Anwar*

Follow My Eyes: Backdoor Attacks on Goal-Directed Scanpath Prediction (Paper) *arXiv 2026*
*Diana Romero, Mutahar Ali, **Momin Ahmad Khan**, Habiba Farrukh, Fatima Muhammad Anwar, Salma Elmalaki*

SABRE-FL: Selective and Accurate Backdoor Rejection for Federated Prompt Learning (Paper) *ICLR 2026*
***Momin Ahmad Khan**, Yasra Chandio, Fatima Muhammad Anwar*

Controlling Vision–Language–Action Policies through Sparse Latent Directions (Paper) *MechInterp @ NeurIPS 2025*
***Momin Ahmad Khan**, Novak Boskov, Fatima Muhammad Anwar, Manzoor Ahmad Khan*

HYDRA-FL: Hybrid Knowledge Distillation for Robust and Accurate Federated Learning (Paper) (Code) *NeurIPS 2024*
***Momin Ahmad Khan**, Yasra Chandio, Fatima Muhammad Anwar*

Decoding FL Defenses: Systemization, Pitfalls, and Remedies (Paper) (Code) *ACM TOPS 2025*
***Momin Ahmad Khan**, Virat Shejwalkar, Yasra Chandio, Amir Houmansadr, Fatima Muhammad Anwar*

Compromising Federated Medical AI-Backdoor Risks in Prompt Learning (Paper) *Poster Abstract @ SenSys 2025*
***Momin Ahmad Khan**, Yasra Chandio, Eugene Bagdasarian, Fatima Muhammad Anwar*

A Neurosymbolic Approach to Adaptive Feature Extraction in SLAM (Paper) *IROS 2024*
*Yasra Chandio, **Momin Ahmad Khan**, Khotso Selialia, Luis Garcia, Joseph DeGol, and Fatima M. Anwar*

On the Pitfalls of Security Evaluation of Robust Federated Learning (Paper) (Code) *DLSP @ IEEE S&P 2023*
***Momin Ahmad Khan**, Virat Shejwalkar, Amir Houmansadr, Fatima Muhammad Anwar*

Security Analysis of SplitFed Learning (Paper) (Code) *AIChallengeIoT @ SenSys 2022*
***Momin Ahmad Khan**, Virat Shejwalkar, Amir Houmansadr, Fatima Muhammad Anwar*

Universal Timestamping with Ambient Sensing (Paper) *SECON 2022*
*Adeel Nasrullah, **Momin Ahmad Khan**, Fatima Muhammad Anwar*

Honors and Awards

- Chancellor's Silver Medal for second-highest EE Department 2021, rank 2/200, NUST
- Rector's Gold Medal for best Undergraduate Final Year Project 2021, NUST
- Recipient of NUST scholarship for outstanding entrance examination score
- NUST High Achievers Award and merit scholarship recipient in 8/8 semesters
- 1st Position in PIEAS Entrance Examination all over Pakistan, 2017
- 1st Position in GIKI Entrance Examination all over Pakistan, 2017

Academic Services

- Reviewer for NeurIPS 2026 (invited), NeurIPS 2025, ICLR 2026, FLCA@AAAI 2026 (invited); invited reviewer for ICLR 2027
- Program Committee member, ACM SenSys 2025 Posters and Demos
- In-person participation in NeurIPS 2024 (Vancouver, Canada), SenSys 2022 (Boston, US), and IEEE S&P 2023 (SF, US)
- Member of organizing team for AI Security Seminar at UMass Amherst
- Member of the organizing team for "AI at the Edge" workshop organized by AI Lounge, Pakistan
- Mentor and project evaluator for SIGMA Lab interns at NUST

Extracurricular Activities

- Runner-up in Dota 2 at NUST E-Sports Gala (2020 & 2021)
- Led door-to-door clothes collection drives for Akhuwat Clothes Bank and distributed them to underserved communities
- Organized donation drive for Master Ayub's School, an open-air school for underprivileged students

Interests and Hobbies

Multiplayer computer games (Top 10% globally in Dota 2), photography, cooking, traveling, playing guitar, and table tennis.